Homework Questions for Molecules and Life Week 1
Due the week of October 27
ANSWER KEY

1. Central Dogma of Molecular Biology, Sickle Cell, and Hemoglobin

In lecture this week, you learned about the Central Dogma of Molecular Biology, which describes the flow of genetic information in all living organisms on our planet. Consider the *HBB* gene, which codes for beta-globin, a subunit of hemoglobin, the protein responsible for transporting oxygen (including the brain, remember fMRI?) and supporting the structure of our red blood cells.

1A. Which of the following options accurately represents the steps involved in the process of converting the information encoded in the *HBB* gene into a functional hemoglobin molecule?

a. HBB gene → HBB mRNA → beta-globin 3D shape → beta-globin amino acid sequence → hemoglobin assembly → hemoglobin function
b. HBB mRNA → HBB DNA → beta-globin amino acid sequence → hemoglobin function → beta-globin 3D shape → hemoglobin assembly
c. HBB DNA → HBB mRNA → hemoglobin function → beta-globin 3D shape → beta-globin amino acid sequence → hemoglobin assembly
d. HBB DNA → HBB mRNA → beta-globin amino acid sequence → beta-globin 3D shape → hemoglobin assembly → hemoglobin function

D is correct

1B. (i) Let's explore the process through which proteins like hemoglobin fold into a specific structure before they can function properly. Protein folding can be understood as a chain of amino acids exploring all of the possible configurations before settling into the correct structure. **How long would it take to fold a hemoglobin protein?** Assume the following and give your answer in Giga-years (1 Gyr = 10^9 years):

- Hemoglobin is 140 amino acids long.
- Each amino acid can have four different configurations.
- It takes half a nanosecond to sample a configuration.

total # of states = $4^{140} = 1.94 \times 10^{84}$ *configurations*

time to sample all states = $(0.5 \times 10^{-9}$ *seconds/configuration*$) \times (1.94 \times 10^{84}$ *configurations*$)$ = 9.7×10^{74} *seconds* *convert to giga-years: 9.7x 10^{74} seconds* $x\,\frac{1\,min}{60\,sec}\,x\,\frac{1\,hr}{60\,min}\,x\,\frac{1\,day}{24\,hr}\,x\,\frac{1\,year}{365\,days}\,x\,\frac{1\,year}{10^9 year}$ *= 3.07 x 10^{58} Giga-years!!!*

(The age of the universe is 13.7 billion years)

1B. (ii) Since we are all alive, proteins **must** fold faster than the previous calculation suggests! Let's instead think about protein folding as forming secondary structures, like alpha helices, first, and then assembling the final protein. **Calculate how long it would take to fold hemoglobin using the number of amino acids above and these new assumptions:**

- A group of 10 amino acids can fold into an alpha helix. Within the alpha helix, each amino acid still has 4 possible configurations and it takes half a nanosecond to sample a configuration.
- These alpha helices form first, and each alpha helix folds independently and in parallel (they fold at the same time).
- Once the alpha helices are folded, groups of alpha helices fold into subunits. There are four subunits of hemoglobin, two subunits contain three alpha helices and two subunits contain four alpha helices. Within each subunit, each alpha helix has six possible configurations and sampling each configuration still takes half a nanosecond. The subunits also fold in parallel. (Hint: only consider the subunit that will take longest to fold)
- Once all four subunits are folded, there is a final series of 20 folding steps to get to the correct final structure. Each of these steps also takes half a nanosecond.

# of states within one alpha helix = 4^{10} = 1.05 x 10^6 configurations

total # of alpha helices = 140/10 = 14 helices

time to sample all configurations to fold an alpha helix = (0.5x10^{-9} seconds/configuration) x (1.05 x 10^6 configurations) = 5.243 x 10^{-4} seconds (this is the time to fold all 14 helices, since they fold in parallel)

To fold the subunits, the limiting step is the subunits with 4 alpha helices:
# of states within a group of 4 helices = 6^4 configurations = 1296 configurations
Time to sample all configurations in a subunit = (1296 configurations) x (0.5x10^{-9} seconds/configuration) = 6.48 x 10^{-7} seconds

time for final folding steps = 20 x 0.5 x 10^{-9} seconds = 1x10^{-8} seconds

total folding time = 5.243 x 10^{-4} seconds + 6.48 x 10^{-7}seconds + 10^{-8} seconds = 5.250 x 10^{-4} seconds

*This is **much** faster than the previous answer, and is in the ballpark of how fast proteins actually fold.*

1C. Mutations to proteins, or changes in the amino acid sequence, can have serious consequences for the function of the protein. In the case of hemoglobin, one specific mutation in the *HBB* gene results in a mutant hemoglobin protein that can clump together, causing red blood cells to form a sickle shape, leading to a disease called sickle cell anemia. Below you are given part of the amino acid sequences from the healthy hemoglobin protein as well as the mutated version of the protein.

Healthy HBB sequence: Leu - Ser - Gly - Glu - Pro - Leu

Mutated HBB sequence: Leu - Ser - Gly - Val - Pro - Leu

An mRNA codon chart (also known as a Genetic Code) is a visual representation that shows the correspondence between mRNA codons (three-nucleotide sequences) and the amino acids they code for during translation. To use an mRNA codon chart, identify the first two bases of the mRNA codon on the chart's left side (First position) and top (Second Position), then use the third base to find the specific amino acid coded by that codon. For amino acids that can be encoded by multiple codons, the codons are grouped together with a bracket. Note that three of the 64 codons code for STOP, meaning that translation ceases with that codon.

Second Position

First Position	U	C	A	G	Third Position
U	UUU, UUC — Phe UUA, UUG — Leu	UCU, UCC, UCA, UCG — Ser	UAU, UAC — Tyr UAA — STOP UAG — STOP	UGU, UGC — Cys UGA — STOP UGG — Trp	U C A G
C	CUU, CUC, CUA, CUG — Leu	CCU, CCC, CCA, CCG — Pro	CAU, CAC — His CAA, CAG — Gln	CGU, CGC, CGA, CGG — Arg	U C A G
A	AUU, AUC, AUA — Ile AUG — Met	ACU, ACC, ACA, ACG — Thr	AAU, AAC — Asn AAA, AAG — Lys	AGU, AGC — Ser AGA, AGG — Arg	U C A G
G	GUU, GUC, GUA, GUG — Val	GCU, GCC, GCA, GCG — Ala	GAU, GAC — Asp GAA, GAG — Glu	GGU, GGC, GGA, GGG — Gly	U C A G

1C. (i) Using the mRNA codon chart provided, fill in the blanks in the table below.

	Healthy	Mutated
Affected amino acid	*Glu (Glutamate)*	*Val (valine)*
Possible mRNA codons	*GAA, GAG*	*GUU, GUC, GUA, GUG*
Complementary DNA codons	*CTT, CTC*	*CAA, CAG, CAT, CAC*

1C. (ii) The mistake you identified above could have occurred in DNA replication, transcription, or translation. **For each process, state which molecular machine would have made the mistake? What would the mistake have been?**

DNA replication: DNA polymerase incorrectly incorporating A instead of T

Transcription: RNA polymerase incorrectly incorporating U instead of A

Translation: ribosome incorrectly incorporating Val instead of Glu

1C (iii) Given that sickle cell anemia is an inheritable disease, which molecular machine must be responsible for the mutation?

In order for the mutation to be inherited, it had to occur in DNA replication

1C. (iv) In the healthy version of hemoglobin, the glutamate amino acid (Glu) keeps individual hemoglobin proteins from interacting with one another. **Using your knowledge of protein sequence and structure, what effect might the mutation you listed in 1B. (i) have on the function of hemoglobin?** Select all that apply.

a. Make it more likely hemoglobin proteins clump together
b. Make it less likely that hemoglobin proteins clump together
c. Have no effect on the hemoglobin protein
d. Make it fold into the correct structure faster

A is correct. The glutamate mutation makes it more likely that the hemoglobin proteins will clump together, forming red blood cells with a sickle shape

1D. In patients with sickle cell anemia, the mutated version of hemoglobin clumps together to form large polymers of hemoglobin that cause red blood cells to sickle. One way to treat the disease is to design a drug that prevents the formation of these large polymers. Researchers have found a compound, GBT440, that binds to the mutated valine residues, increasing the hemoglobin's ability to bind oxygen and decreasing its ability to form large polymers. The researchers determined the structure of the compound bound to hemoglobin, shown in Figure 1 on the next page.

1D. (i) Which technique(s) could the researchers have used to generate the structure in Figure 1, and what type of radiation is used in that technique?

a. X-ray crystallography, using x-rays
b. Cryo-EM, using electron beams
c. Fluorescent light microscopy, using visible light
d. Cryo-EM, using visible light

A & B is correct

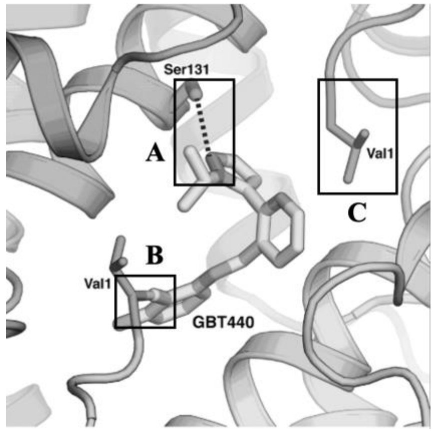

Figure 1. *Structure of GBT440 bound to hemoglobin.*

1D. (ii) Based on your answer to question 1B (ii), why might the researchers have been excited to see that GBT440 binds to the mutated valine residue?

a. It indicates that GBT440 enhances the sickling of red blood cells.

b. Binding to the mutated valine prevents hemoglobin proteins from sticking together, reducing polymer formation and sickling of red blood cells.

c. GBT440 binding to valine encourages new mutations, facilitating more protein interactions.

d. It suggests that GBT440 can bind to other forms of hemoglobin, improving oxygen delivery to tissues.

B is correct. Binding to the mutated valine prevents hemoglobin proteins from sticking together, reducing polymer formation and sickling of red blood cells.

1E. The researchers needed to test the effects of GBT440 on the sickling of red blood cells in the laboratory before they even thought about giving it to patients! To do this, they added different amounts of GBT440 to samples of blood from sickle cell patients and measured the amount of red blood cells that had the sickled shape. The results are shown in Figure 2 on the next page.

1E. (i) Based on the results in Figure 2, does GBT440 reduce the amount of red blood cells that are sickled? Explain using your knowledge of statistics.

Yes, GBT440 at even 300ug/L significantly reduces the percentage of sickled red blood cells compared to the percentage with no GBT440 added. This difference is significant because the p-value is 0.03, less than our standard threshold of 0.05.

1E. (ii) Based on the results in Figure 2, does increasing the amount of GBT440 change its effect on the sickling of red blood cells? Explain using your knowledge of statistics.

No, there is not a significant difference between the percentage of sickled red blood cells with 300 and 600 ug/L of GBT440, shown by the p-value of 0.19 (greater than 0.05). At higher doses, the only sickled red blood cells left are those that are irreversibly sickled.

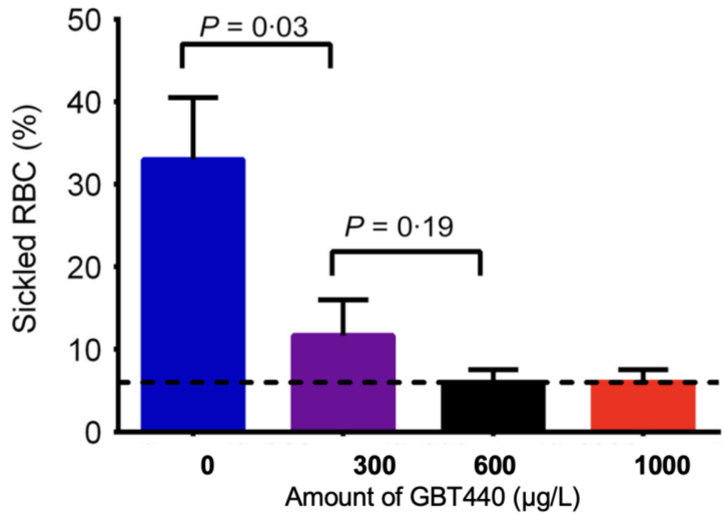

Figure 2. *Percentage of red blood cells (RBC) that had a sickle shape after the addition of various amounts of GBT440. The p-values for select comparisons are listed above the corresponding brackets. The dotted line indicates the percentage of irreversibly sickled RBCs.*

1F. (i) The researchers then moved on to clinical trials with GBT440. In this study, they gave the patients GBT440 over the course of 90 days, collected samples of the patients' blood, and monitored the percentage of sickled red blood cells over time. The results are shown in Figure 3 below.

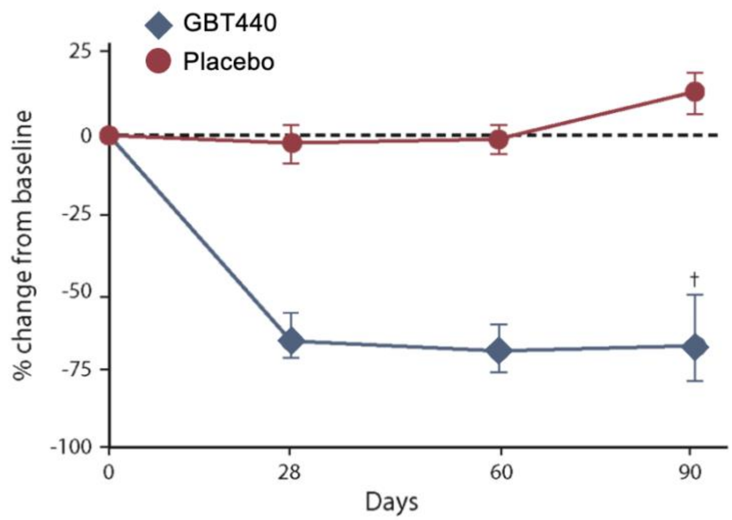

Figure 3. *Percentage change from baseline level of sickled red blood cells in patients given GBT440 compared to those given a placebo containing all the same components as the GBT440 treatment except the compound itself. Error bars represent the 95% confidence intervals.*

1F (ii). Based on Figure 3, is GBT440 effective at reducing the amount of sickled red blood cells in patients? Explain using your knowledge of statistics.

Yes, the GBT440 treatment significantly reduces the amount of sickled red blood cells compared to baseline and compared to the patients receiving the placebo. The difference is significant *because the 95% confidence intervals don't overlap.*

1G. How do genetic mutations lead to observable diseases like sickle cell anemia, and what does this suggest about the potential for targeted medical treatments in the future?

a. Genetic mutations can change how proteins function, potentially causing disease symptoms; this knowledge supports the development of treatments tailored to mutations individual patients have.

b. Most genetic mutations are harmless, so they are unlikely to contribute to diseases or be useful targets for treatment.

c. Genetic mutations typically have no impact on protein function, making disease symptoms unpredictable.

d. Mutations affect DNA, but because the body can often compensate at the cellular level, they rarely lead to symptoms that can be medically targeted.

A is correct. Genetic mutations may alter proper protein function and can lead to observable disease symptoms. Understanding these mutations allows for the development of targeted treatments and underscores the promise of personalized medicine to address an individual's specific mutations more effectively.

2. <u>Nucleotide probabilities</u>

The average human cell experiences 50,000 DNA damage events per day. When these occur in coding regions, they have the potential to cause harmful changes to protein structure and function. Let's explore the consequences of this using the example of hemoglobin.

2A. (i) The human genome is roughly 3.1 billion nucleotides long. If a DNA damage event causes a single mutation, what are the odds that this mutation will occur within the nucleotide sequence that codes for the hemoglobin protein's 140 amino acids?

*A 140 amino-acid protein includes 420 (140 * 3) nucleic acids in its coding region. The odds of a single event falling in that region are thus:*

$$\frac{420}{3.1 * 10^{9}} = 1.35 * 10^{-7}$$

2A. (ii) What are the odds that NONE of the 50,000 daily DNA damage events will affect the *HBB* gene?

$(1 - 1.35 * 10^{-7})^{50,000} = 0.993$

2A. (iii) What are the odds that at least ONE of the daily DNA damage events will affect the *HBB* gene? Hint: P(at least once) = 1 – P(never)

The probability that at least one of the 50,000 events occurs in the hemoglobin gene are:

$$1 - 0.993 = 7 * 10^{-3}$$

2A. (iv) What are the odds that the *HBB* gene in a given cell will experience damage over the course of one year?

This is the probability of a mutation not happening once a day in 365 tries:

$$1 - ((1 - 7 * 10^{-3})^{365}) = 0.923$$

A remarkably high probability! And with 30 trillion cells in the body, the importance of DNA repair becomes clear.